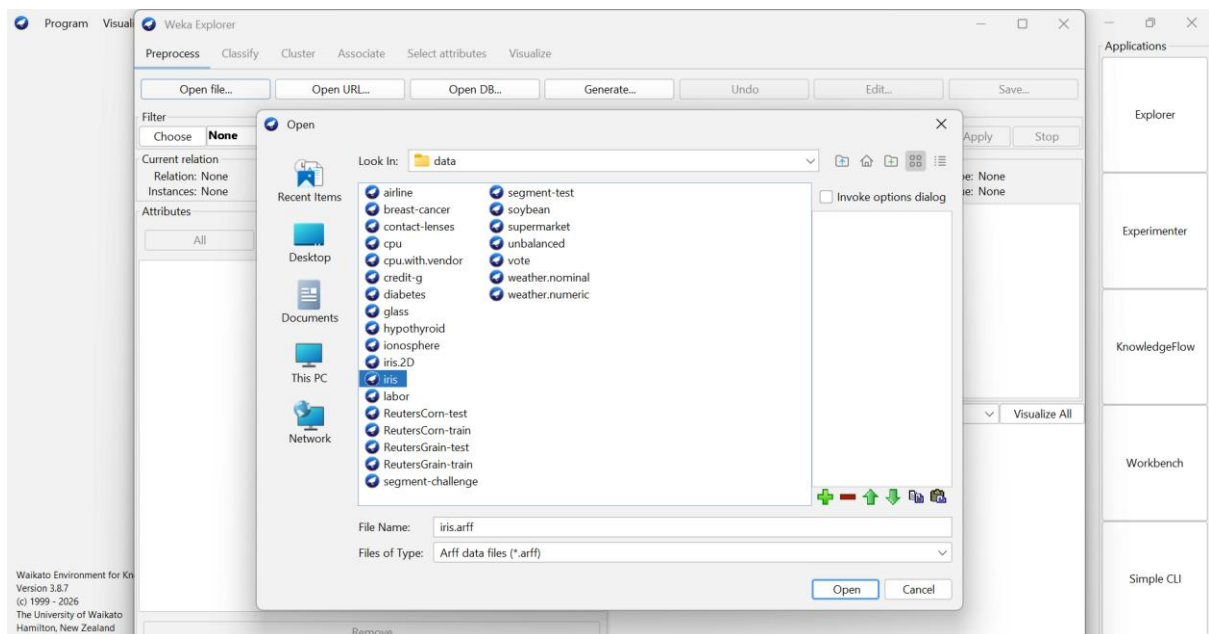
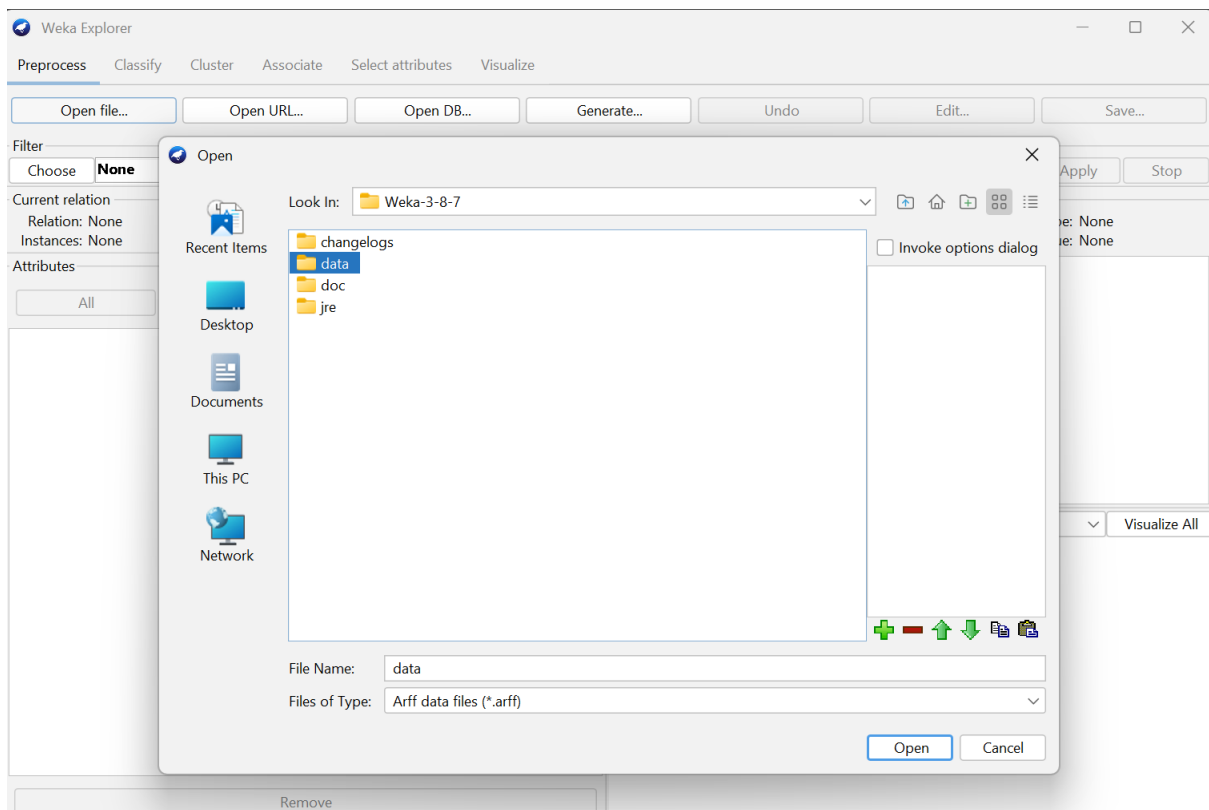


EXPERIMENT-28



Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **J48 -C 0.25 -M 2**

Test options:
☐ Use training set
☐ Supplied test set (Set...)
☒ Cross-validation Folds **10**
☐ Percentage split % **66**
 More options...

(Nom) class: Start Stop

Result list (right-click for options):
 21:18:26 - trees.J48

Classifier output:

```

=== Run information ===

Scheme:      weka.classifiers.trees.J48 -C 0.25 -M 2
Relation:    pima_diabetes
Instances:   768
Attributes:  9
  preg
  plas
  pres
  skin
  insu
  mass
  pedi
  age
  class

Test mode:   10-fold cross-validation

=== Classifier model (full training set) ===

J48 pruned tree

-----

plas <= 127
| mass <= 26.4: tested_negative (132.0/3.0)
| mass > 26.4
| | age <= 28: tested_negative (180.0/22.0)
| | age > 28
| | | plas <= 99: tested_negative (55.0/10.0)
| | | plas > 99

```

Status: OK Log x 0

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **J48 -C 0.25 -M 2**

Test options:
☐ Use training set
☐ Supplied test set (Set...)
☒ Cross-validation Folds **10**
☐ Percentage split % **66**
 More options...

(Nom) class: Start Stop

Result list (right-click for options):
 21:18:26 - trees.J48

Classifier output:

```

plas <= 127
| mass <= 26.4: tested_negative (132.0/3.0)
| mass > 26.4
| | age <= 28: tested_negative (180.0/22.0)
| | age > 28
| | | plas <= 99: tested_negative (55.0/10.0)
| | | plas > 99
| | | | pedi <= 0.56: tested_negative (84.0/34.0)
| | | | pedi > 0.56
| | | | | preg <= 6
| | | | | age <= 30: tested_positive (4.0)
| | | | | age > 30
| | | | | | age <= 34: tested_negative (7.0/1.0)
| | | | | | age > 34
| | | | | | | mass <= 33.1: tested_positive (6.0)
| | | | | | | mass > 33.1: tested_negative (4.0/1.0)
| | | | | | | preg > 6: tested_positive (13.0)
plas > 127
| mass <= 29.9
| | plas <= 145: tested_negative (41.0/6.0)
| | plas > 145
| | | age <= 25: tested_negative (4.0)
| | | age > 25
| | | | age <= 61
| | | | | mass <= 27.1: tested_positive (12.0/1.0)
| | | | | mass > 27.1
| | | | | | pres <= 82
| | | | | | | pedi <= 0.396: tested_positive (8.0/1.0)
| | | | | | | pedi > 0.396: tested_negative (3.0)

```

Status: OK Log x 0

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **J48 -C 0.25 -M 2**

Test options:

- ☐ Use training set
- ☐ Supplied test set
- ☒ Cross-validation Folds **10**
- ☐ Percentage split % **66**

(Nom) class

Result list (right-click for options)

- 21:18:26 - trees.J48

Classifier output:

Time taken to build model: 0.05 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	567	73.8281 %
Incorrectly Classified Instances	201	26.1719 %
Kappa statistic	0.4164	
Mean absolute error	0.3158	
Root mean squared error	0.4463	
Relative absolute error	69.4841 %	
Root relative squared error	93.6293 %	
Total Number of Instances	768	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
Weighted Avg.	0.814	0.403	0.790	0.814	0.802	0.417	0.751	0.811	tested_negative
	0.597	0.186	0.632	0.597	0.614	0.417	0.751	0.572	tested_positive

=== Confusion Matrix ===

	a	b	<-- classified as
407 93	a = tested_negative		
108 160	b = tested_positive		

Status: OK  x 0

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **SMO -C 1.0 -L 0.001 -P 1.0E-12 -N 0 -V -1 -W 1 -K "weka.classifiers.functions.supportVector.PolyKernel -E 1.0 -C 250007" -calibrator "weka.classifiers.functions.Logistic -R 1.0E-8 -M -1 -num-decimal-places 4"**

Test options:

- ☐ Use training set
- ☐ Supplied test set
- ☒ Cross-validation Folds **10**
- ☐ Percentage split % **66**

(Nom) class

Result list (right-click for options)

- 21:18:26 - trees.J48
- 21:22:08 - functions.SMO

Classifier output:

=== Run information ===

Scheme: weka.classifiers.functions.SMO -C 1.0 -L 0.001 -P 1.0E-12 -N 0 -V -1 -W 1 -K "weka.classifiers.functions.supportVector.PolyKernel -E 1.0 -C 250007" -calibrator "weka.classifiers.functions.Logistic -R 1.0E-8 -M -1 -num-decimal-places 4"

Relation: pima_diabetes

Instances: 768

Attributes: 9

Test mode: 10-fold cross-validation

=== Classifier model (full training set) ===

SMO

Kernel used:

Linear Kernel: $K(x, y) = \langle x, y \rangle$

Classifier for classes: tested_negative, tested_positive

BinarySMO

Status: OK  x 0

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **SMO** -C 1.0 -L 0.001 -P 1.0E-12 -N 0 -V -1 -W 1 -K "weka.classifiers.functions.supportVector.PolyKernel -E 1.0 -C 250007" -calibrator "weka.classifiers.functions.Logistic -R 1.0E-8 -M -1 -num-decimal-places 4"

Test options:
☐ Use training set
☐ Supplied test set Set...
☒ Cross-validation Folds **10**
☐ Percentage split % **66**
 More options...

(Nom) class
 Start Stop
 Result list (right-click for options):
 21:18:26 - trees.J48
21:22:08 - functions.SMO

Classifier output
 BinarySMO
 Machine linear: showing attribute weights, not support vectors.

```

1.3614 * (normalized) preg
+
4.8764 * (normalized) plas
+
-0.8118 * (normalized) pres
+
-0.1158 * (normalized) skin
+
-0.1776 * (normalized) insu
+
3.0745 * (normalized) mass
+
1.4242 * (normalized) pedi
+
0.2601 * (normalized) age
-
5.1761

```

 Number of kernel evaluations: 19131 (69.279% cached)
 Time taken to build model: 0.06 seconds
 === Stratified cross-validation ===
 === Summary ===

Correctly Classified Instances	594	77.3438 %
Incorrectly Classified Instances	174	22.6563 %
Kappa statistic	0.4682	
Mean absolute error	0.2266	
Root mean squared error	0.476	
Relative absolute error	49.848 %	

Status: OK Log

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **SMO** -C 1.0 -L 0.001 -P 1.0E-12 -N 0 -V -1 -W 1 -K "weka.classifiers.functions.supportVector.PolyKernel -E 1.0 -C 250007" -calibrator "weka.classifiers.functions.Logistic -R 1.0E-8 -M -1 -num-decimal-places 4"

Test options:
☐ Use training set
☐ Supplied test set Set...
☒ Cross-validation Folds **10**
☐ Percentage split % **66**
 More options...

(Nom) class
 Start Stop
 Result list (right-click for options):
 21:18:26 - trees.J48
21:22:08 - functions.SMO

Classifier output
 Time taken to build model: 0.06 seconds
 === Stratified cross-validation ===
 === Summary ===

Correctly Classified Instances	594	77.3438 %
Incorrectly Classified Instances	174	22.6563 %
Kappa statistic	0.4682	
Mean absolute error	0.2266	
Root mean squared error	0.476	
Relative absolute error	49.848 %	
Root relative squared error	99.862 %	
Total Number of Instances	768	

 === Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.898	0.459	0.785	0.898	0.838	0.480	0.720	0.771	tested_negative
	0.541	0.102	0.740	0.541	0.625	0.480	0.720	0.560	tested_positive
Weighted Avg.	0.773	0.334	0.769	0.773	0.763	0.480	0.720	0.698	

 === Confusion Matrix ===

```

a b <-- classified as
449 51 | a = tested_negative
123 145 | b = tested_positive

```

Status: OK Log